```
players = ["Serena Williams", "Lionel Messi", "Michael Phelps",
           "Usain Bolt", "Roger Federer", "Cristiano Ronaldo"]

sports = ["Tennis", "Soccer", "Swimming", "Athletics", "Tennis", "Soccer"]

achievements = [23, 7, 23, 8, 20, 5]

for i in range(len(players)):
    if players[i] == "Lionel Messi":
        if achievements[i] > 10:
            print("Player Name:", players[i])
            print("Sport:", sports[i])
            print("Achievements:", achievements[i])
        else:
            print("Lionel Messi does not have more than 10 achievements")
```

```
Lionel Messi does not have more than 10 achievements
```

```
#question2

players = ["Serena Williams", "Lionel Messi", "Michael Phelps",
           "Usain Bolt", "Roger Federer", "Cristiano Ronaldo"]

sports = ["Tennis", "Soccer", "Swimming", "Athletics", "Tennis", "Soccer"]

achievements = [23, 7, 23, 8, 20, 5]

for i in range(len(players)):
    if sports[i] == "Tennis" or achievements[i] == 20:
        print("Player Name:", players[i])
        print("Sport:", sports[i])
        print("Achievements:", achievements[i])
```

```
Player Name: Serena Williams
Sport: Tennis
Achievements: 23
Player Name: Roger Federer
Sport: Tennis
Achievements: 20
```

```
#question 3
for i in range(len(players)):
    if achievements[i] < 10 and sports[i] != "Soccer":
        print("Player Name:", players[i])
        print("Sport:", sports[i])
        print("Achievements:", achievements[i])
```

```
Player Name: Usain Bolt
Sport: Athletics
Achievements: 8
```

```
Start coding or generate with AI.
```

```
#question 4

num = int(input("Enter an integer between 1 and 9999: "))

# Even or Odd
if num % 2 == 0:
    print("Even")
else:
    print("Odd")

# Palindrome
if str(num) == str(num)[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")

# Sum of digits
temp = num
sum_digits = 0

while temp > 0:
```

```
    digit = temp % 10
    sum_digits = sum_digits + digit
    temp = temp // 10

print("Sum of digits:", sum_digits)

# Divisible by sum of digits
if num % sum_digits == 0:
    print("Divisible by the sum of digits")
else:
    print("Not divisible by the sum of digits")
```

```
Enter an integer between 1 and 9999: 1221
Odd
Palindrome
Sum of digits: 6
Not divisible by the sum of digits
```

```
#question 5

for i in range(-5, 6):
    print(i)
```

```
-5
-4
-3
-2
-1
0
1
2
3
4
5
```

```
#question 6

squares = ['red', 'yellow', 'green', 'purple', 'blue']
```

```
for i in squares:
    print(i)
```

```
red
yellow
green
purple
blue
```

```
new_squares = []

i = 0

while i < len(squares):
    if squares[i] != 'orange':
        break

    new_squares.append(squares[i])
    i = i + 1

print(new_squares)
```

```
[]
```

```
animals = ["lion", "giraffe", "gorilla", "parrots", "crocodile", "deer", "swan"]

new_animals = []

i = 0

while i < len(animals):
    if len(animals[i]) == 7:
        new_animals.append(animals[i])

    i = i + 1

print(new_animals)
```

```
['giraffe', 'gorilla', 'parrots']
```

```
Start coding or generate with AI.
```